

DBP Prediction

Ablation Study & SHAP Analysis

Incremental Feature Selection for T-THMs, DBCM, and BDCM

Models: Random Forest | XGBoost | MLP | KAN

Group Meeting Report — March 2026

1 Study Overview

Motivation, dataset & experimental methodology

Objective

Systematically evaluate the contribution of each water quality parameter to DBP (Disinfection By-Product) prediction by incrementally adding features and tracking both model performance (R^2 , RMSE) and feature importance (SHAP values) across four model families.

Dataset

Property	Value
Source	DWTP-B (single water treatment)
Total samples	175 (141 train / 34 test —
Targets	T-THMs • DBCM (chlorinated) •
All 9 features	pH, UV254, Temperature, TOC, C
Evaluation	Test R^2 , Test RMSE, 5-fold C

Ablation Rounds — Incremental Feature Addition

Round	Feature Set	Added Feature	Rationale
R1 (3-feat)	pH + UV254 + Temperature	—	Core water-quality indicators
R2 (4-feat)	+ TOC	+ TOC	Dissolved organic carbon
R3 (5-feat)	+ COD	+ COD	Organic demand / reactivity
R4 (6-feat)	+ Br	+ Br	Bromine precursor for BDCM
R5 (9-feat)	+ $\text{NH}_4\text{-N}$ + $\text{NO}_2\text{-N}$ + $\text{NO}_3\text{-N}$	+ Nitrogen	Chloramine formation pathway

Methodology

- Per-round pipeline: Scout run (3-fold CV, ~30 trials) -> Formal tuning (5-fold CV, 50-80 trials)
- Tuner: Optuna TPE sampler with median pruning; stability penalty $\lambda = 0.18$
- Strategy: Per-target — independent model for each of the 3 DBP targets
- Interpretability: SHAP (TreeExplainer for RF/XGB; KernelExplainer for MLP/KAN)

2 Ablation Performance Summary

Best-model test R^2 and RMSE per target across feature rounds

Target: T-THMs (Total Trihalomethanes)

Feature Set	Best Model	R^2	RMSE	ΔR^2 vs Prev
3-feat	RF	0.545	6.90	—
4-feat (+TOC)	RF	0.549	6.86	+0.004
5-feat (+COD)	RF	0.571	6.70	+0.022 (*)
6-feat (+Br)	RF	0.492	7.29	-0.079
9-feat (+N)	XGB	0.533	6.98	+0.041

Target: DBCM (Dibromochloromethane)

Feature Set	Best Model	R^2	RMSE	ΔR^2 vs Prev
3-feat	MLP	0.597	2.46	—
4-feat (+TOC)	MLP	0.638	2.33	+0.041
5-feat (+COD)	MLP	0.659	2.27	+0.021 (*)
6-feat (+Br)	KAN	0.566	2.56	-0.093
9-feat (+N)	XGB	0.561	2.57	-0.005

Target: BDCM (Bromodichloromethane)

Feature Set	Best Model	R^2	RMSE	ΔR^2 vs Prev
3-feat	XGB	0.314	1.72	—
4-feat (+TOC)	RF	0.386	1.62	+0.072
5-feat (+COD)	RF	0.393	1.61	+0.007
6-feat (+Br)	MLP	0.434	1.56	+0.041 (*)
9-feat (+N)	RF	0.391	1.62	-0.043

Key Takeaway

- Optimal feature count: 5 features for T-THMs and DBCM; 6 features (+ Br) for BDCM.
- Nitrogen indicators: $\text{NH}_4\text{-N}$, $\text{NO}_2\text{-N}$, $\text{NO}_3\text{-N}$ add no performance gain on this dataset.
- 6-feature regression: Adding Br hurts T-THMs/DBCM — Bromine acts as noise for chlorinated DBPs.

3 SHAP — Best Model per Target

T-THMs: Random Forest with 5 features

Feature Importance Bar Chart

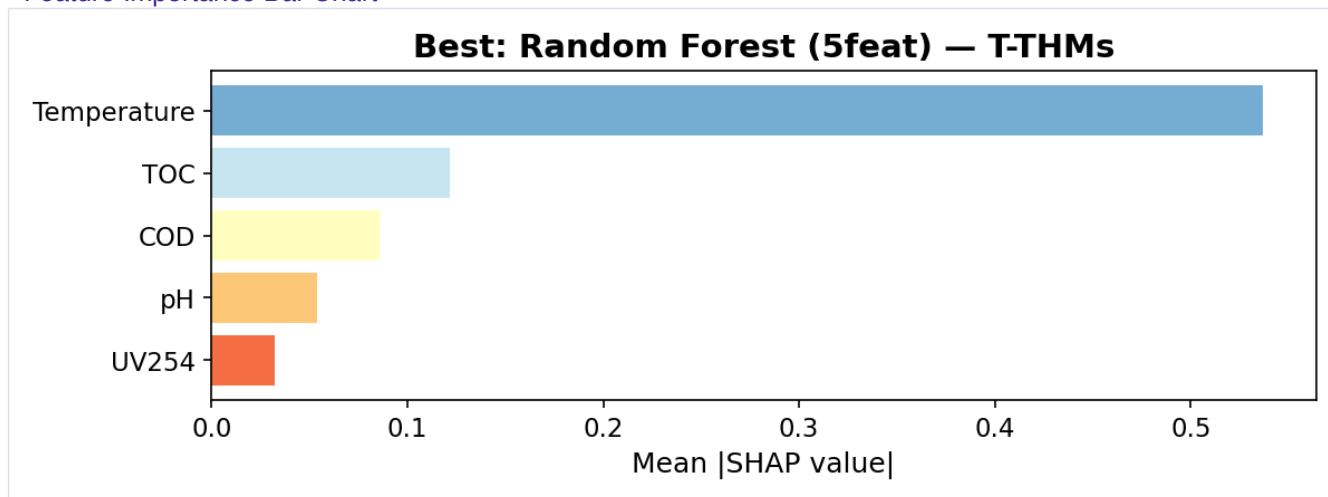


Fig 1a Mean |SHAP| values — T-THMs (RF, 5-feat)

SHAP Beeswarm Summary

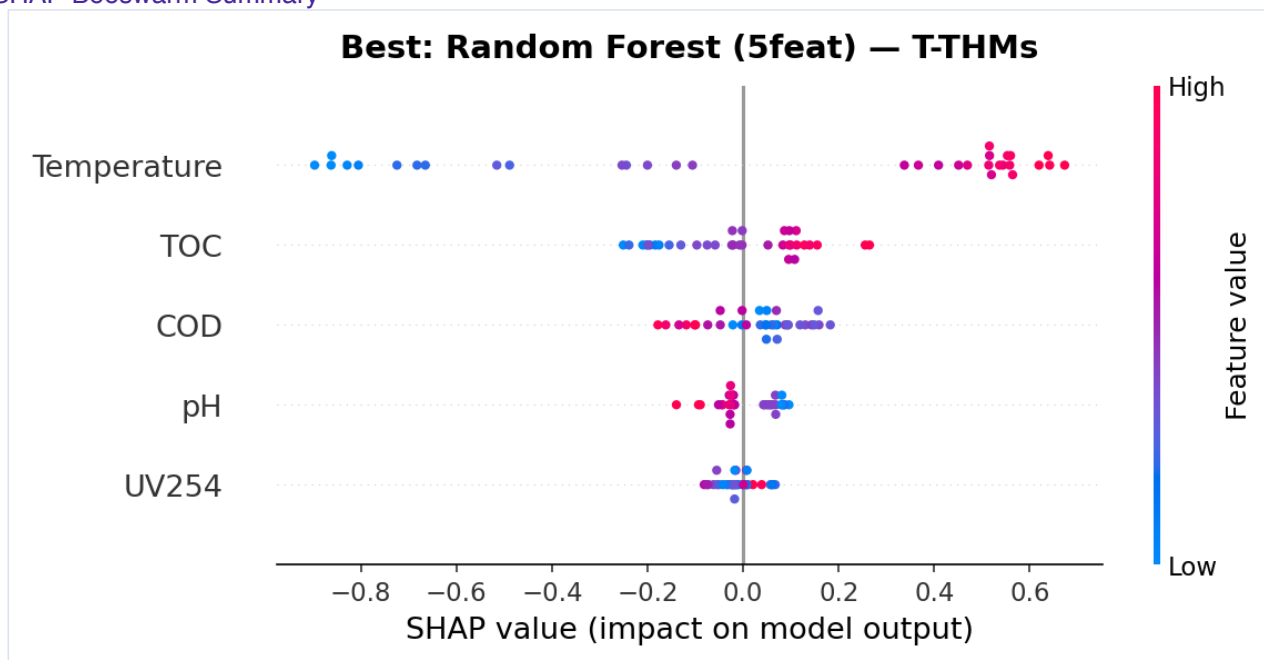


Fig 1b SHAP beeswarm: feature value vs impact on prediction — T-THMs

3 SHAP — Best Model per Target (cont.)

DBCM: MLP with 5 features

Feature Importance Bar Chart

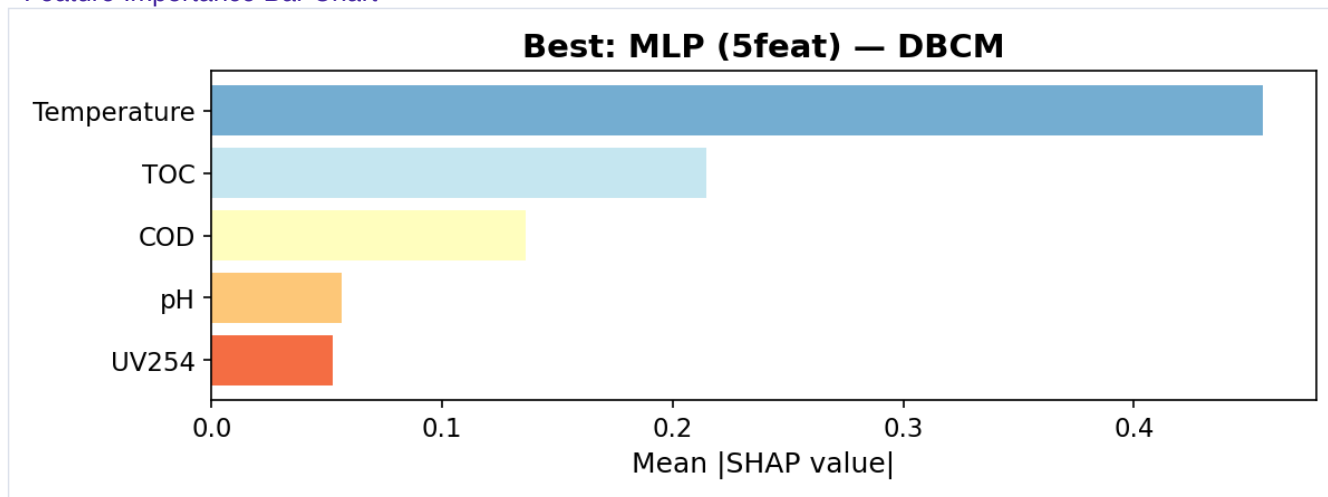


Fig 2a Mean SHAPI values — DBCM (MLP, 5-feat)

SHAP Beeswarm Summary

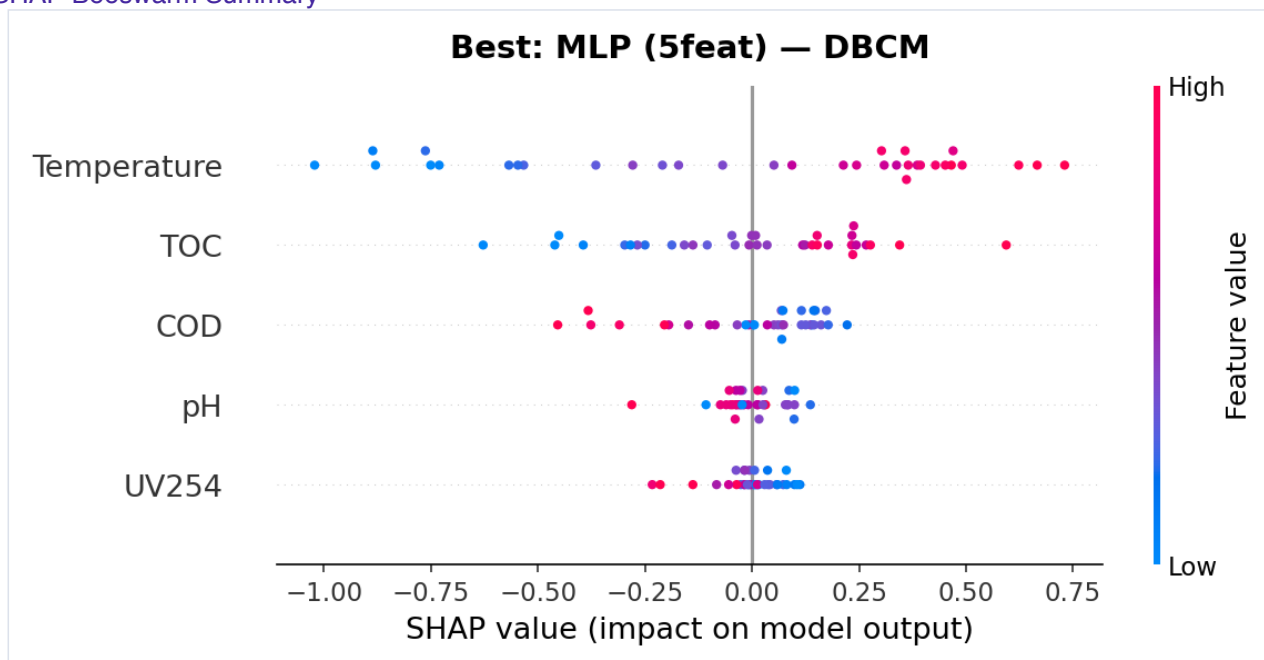


Fig 2b SHAP beeswarm — DBCM (MLP, 5-feat)

3 SHAP — Best Model per Target (cont.)

BDCM: MLP with 6 features (includes Br⁻)

Feature Importance Bar Chart

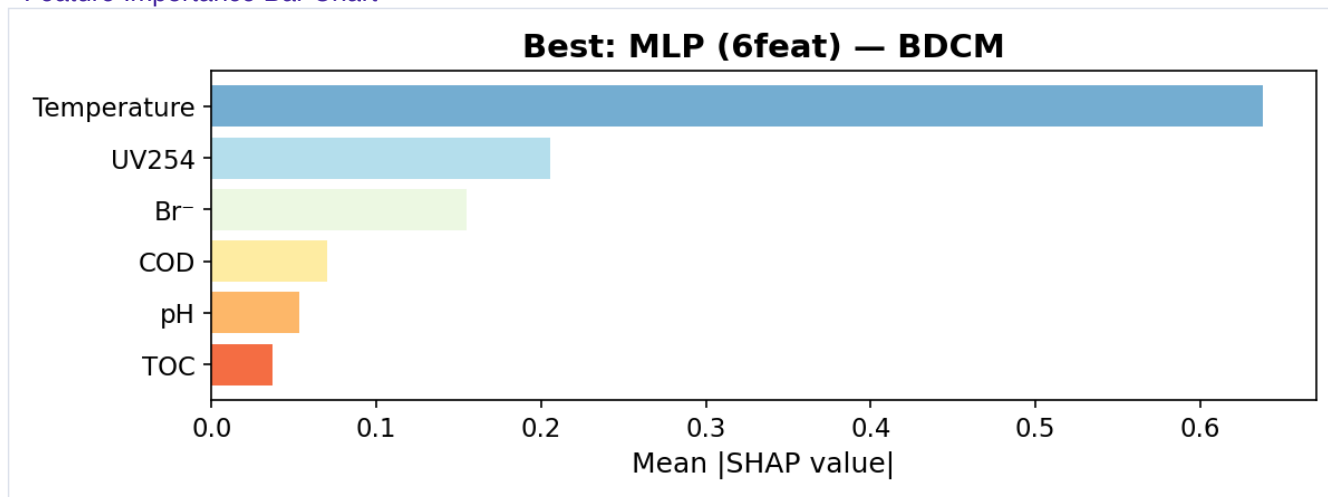


Fig 3a Mean |SHAP| values — BDCM (MLP, 6-feat)

SHAP Beeswarm Summary

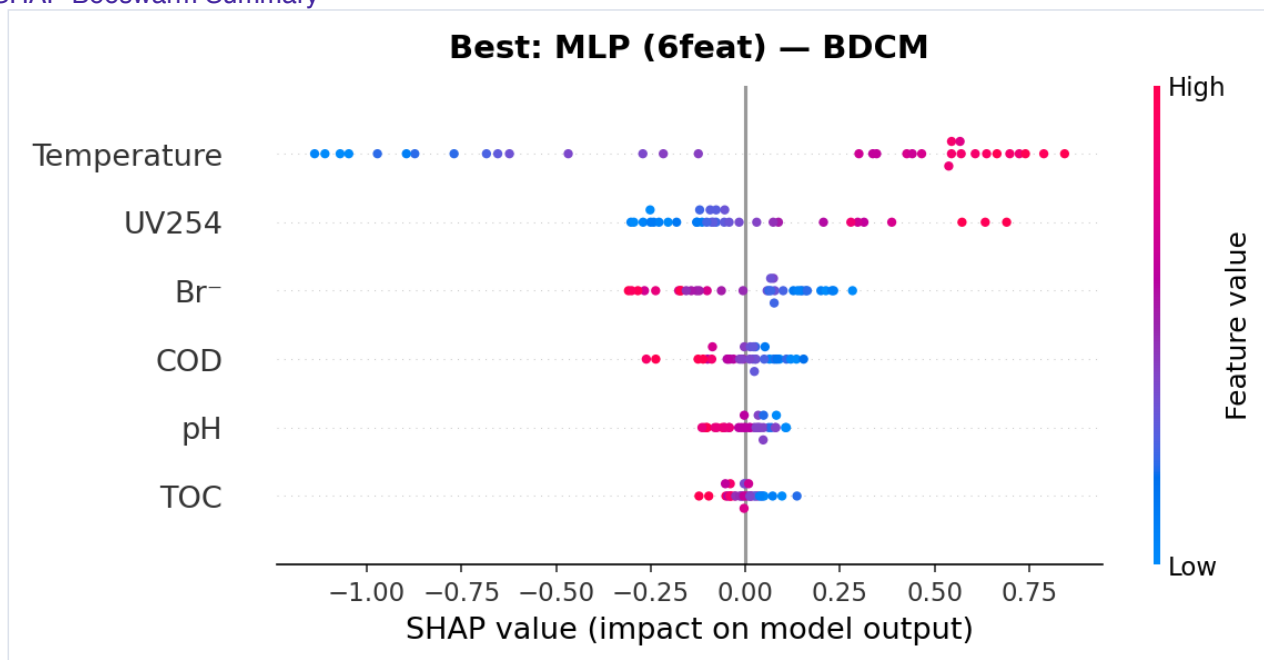


Fig 3b SHAP beeswarm — BDCM (MLP, 6-feat)

3 SHAP Dependence Plots

Key feature-target non-linear interactions

T-THMs × Temperature

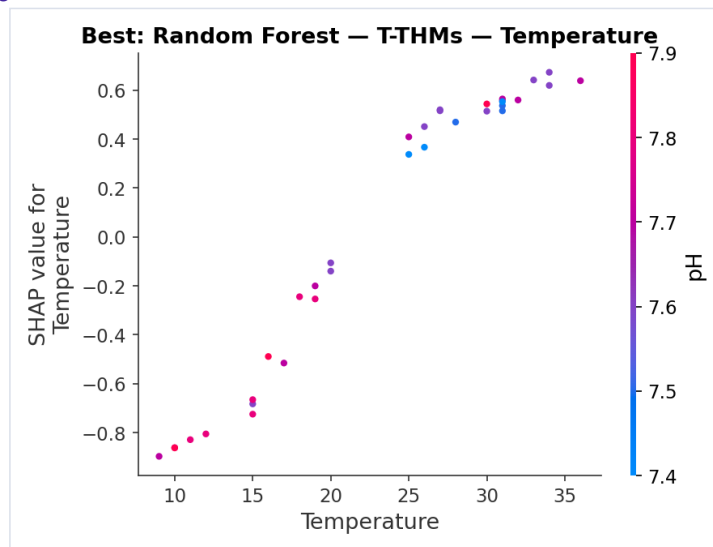


Fig 4a Temperature vs SHAP impact on T-THMs (RF, 5-feat)

DBCM × Temperature

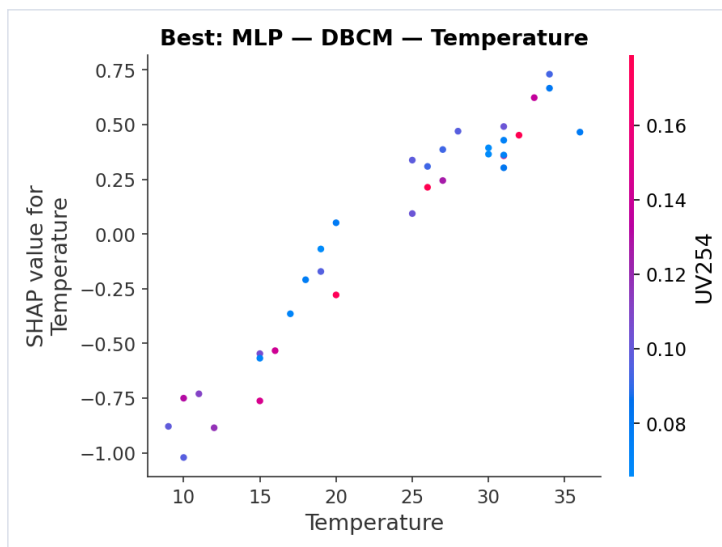


Fig 4b Temperature vs SHAP impact on DBCM (MLP, 5-feat)

BDCM × Temperature

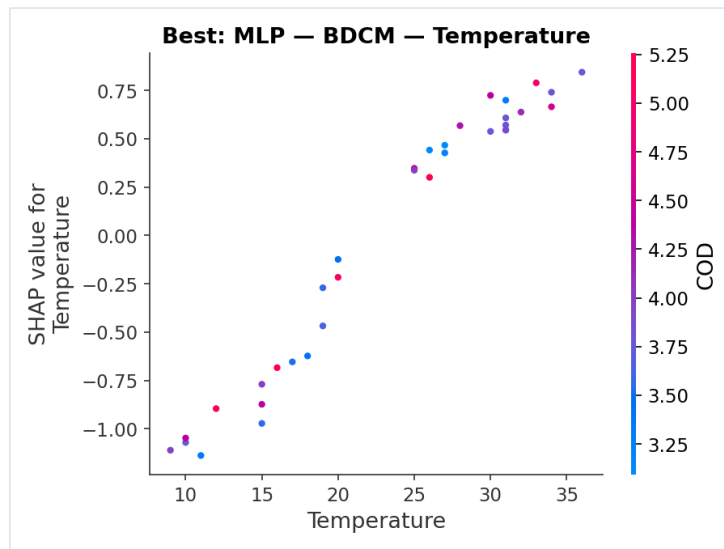


Fig 4c Temperature vs SHAP impact on BDCM (MLP, 6-feat)

4 Cross-Model Feature Importance (9 Features)

Consensus across RF, XGBoost, MLP, and KAN

Target: T-THMs

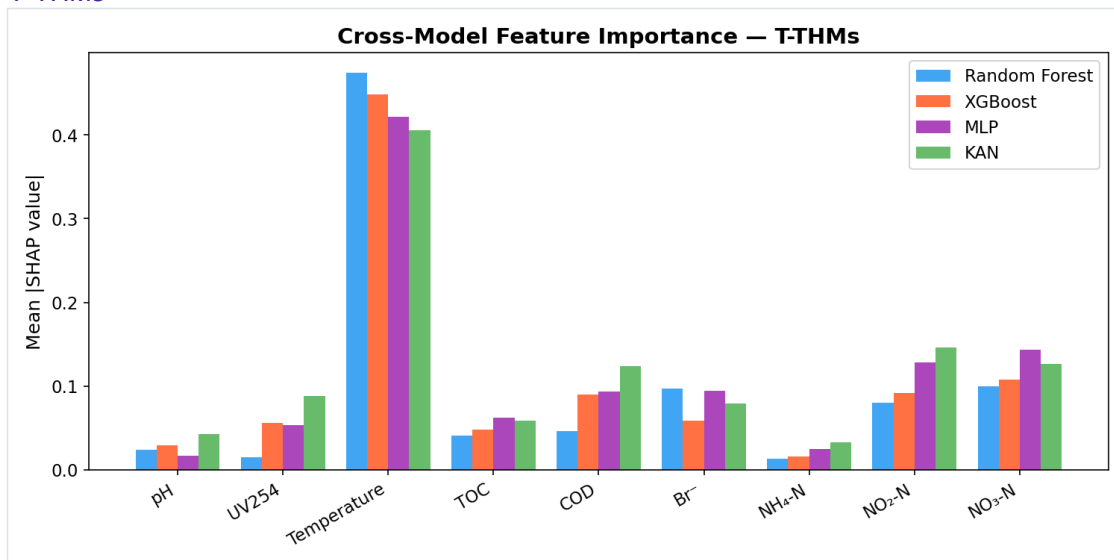


Fig 5a All-model SHAP importance — T-THMs (9-feat)

Target: DBCM

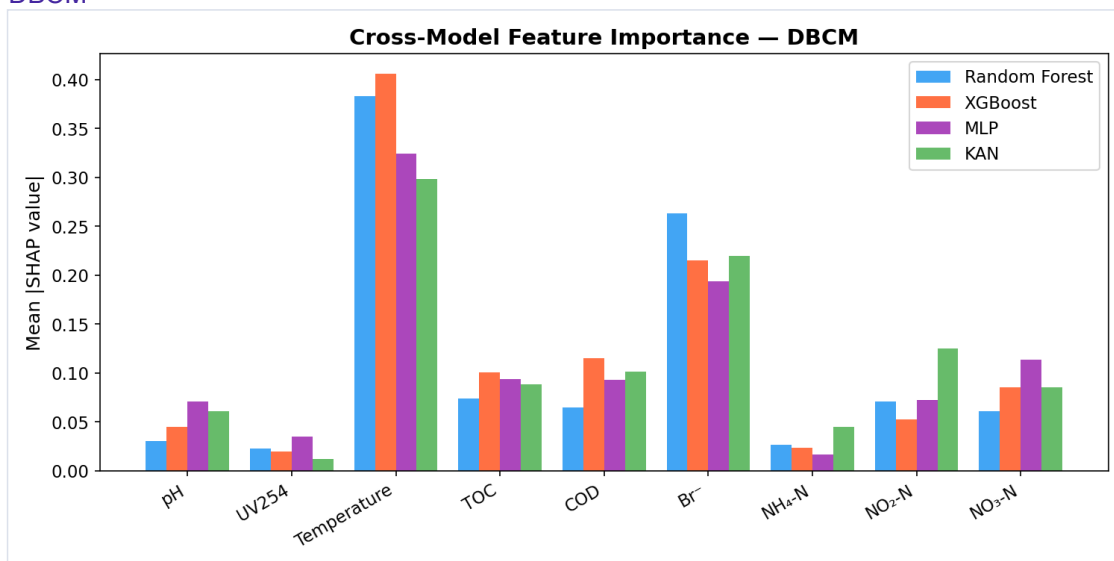


Fig 5b All-model SHAP importance — DBCM (9-feat)

Target: BDCM

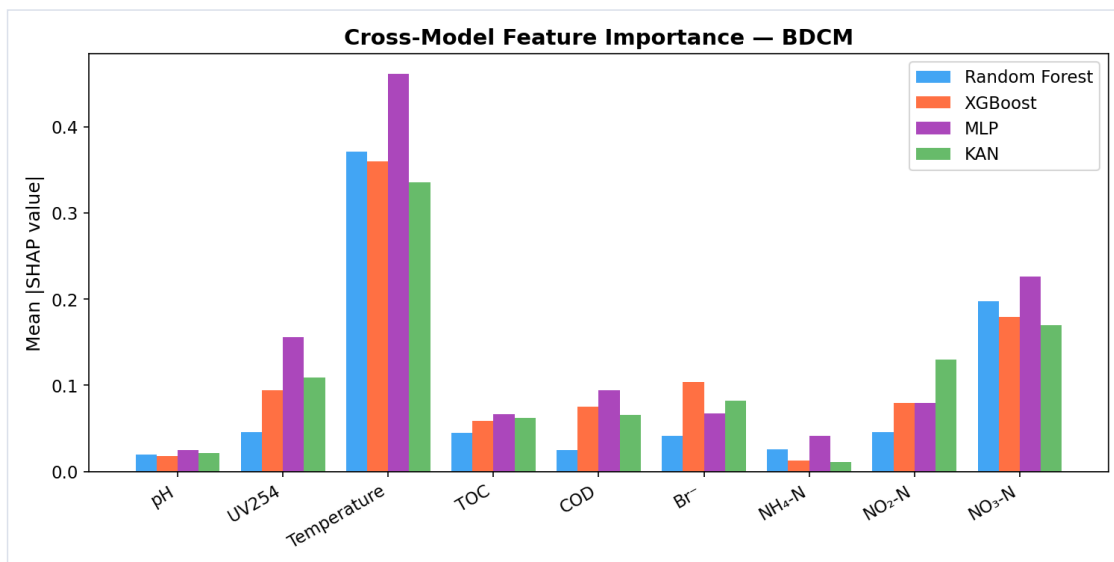


Fig 5c All-model SHAP importance — BDCM (9-feat)

5 Feature Importance Evolution — Random Forest

Mean ISHAPI per feature as ablation round increases from 3 to 9

Target: T-THMs

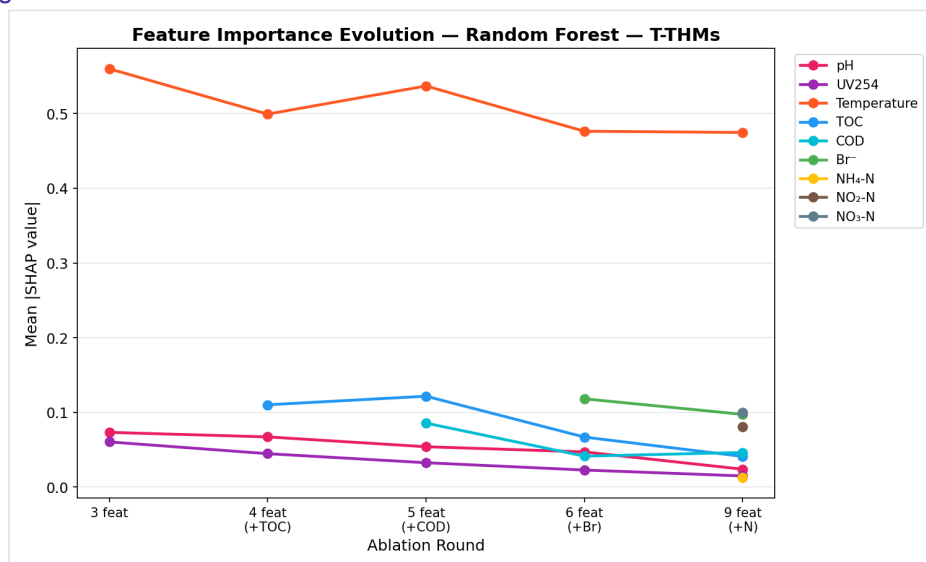


Fig 6a RF importance evolution — T-THMs

Target: DBCM

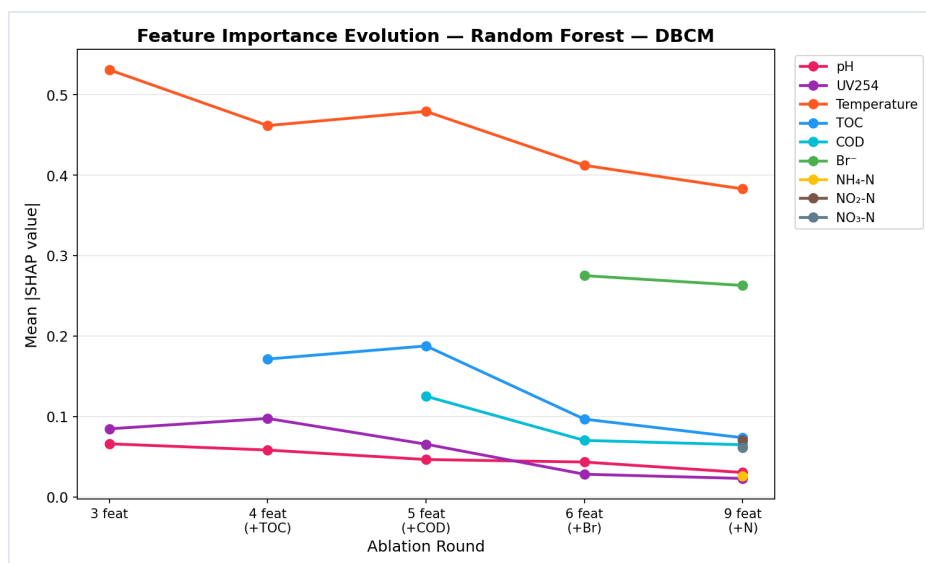


Fig 6b RF importance evolution — DBCM

Target: BDCM

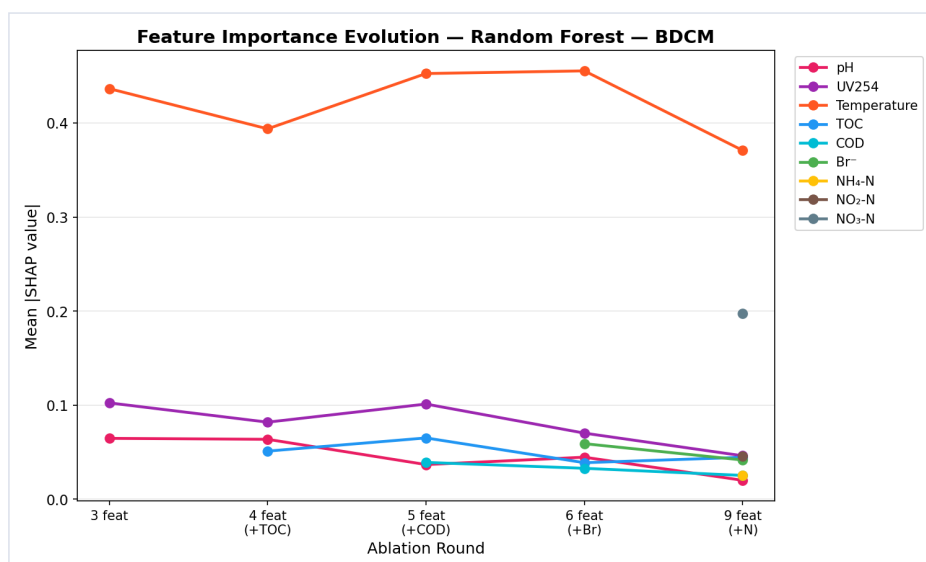


Fig 6c: RF importance evolution — BDCM

5 Feature Importance Evolution — XGBoost

Consistent patterns with Random Forest; confirms cross-model consensus

Target: T-THMs

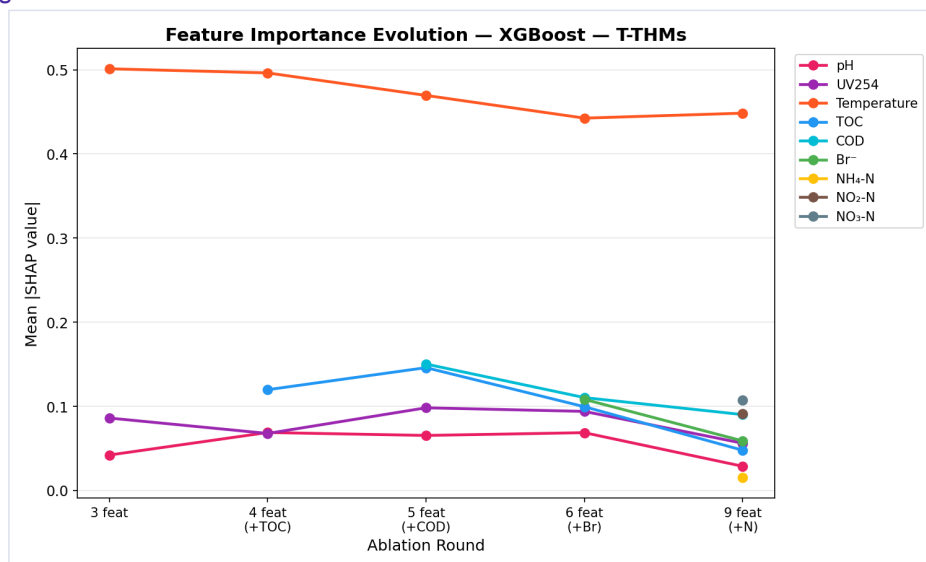


Fig 7a XGBoost importance evolution — T-THMs

Target: DBCM

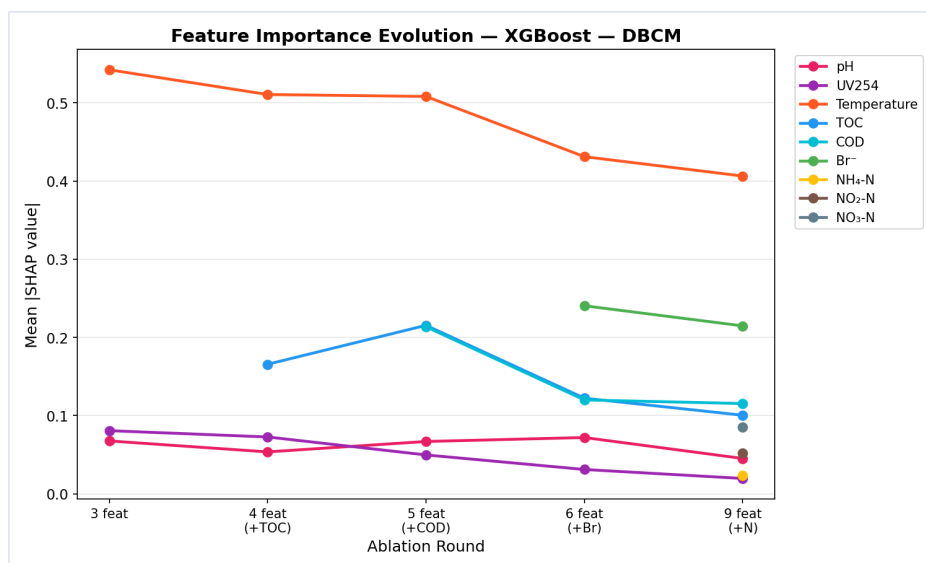


Fig 7b XGBoost importance evolution — DBCM

Target: BDCM

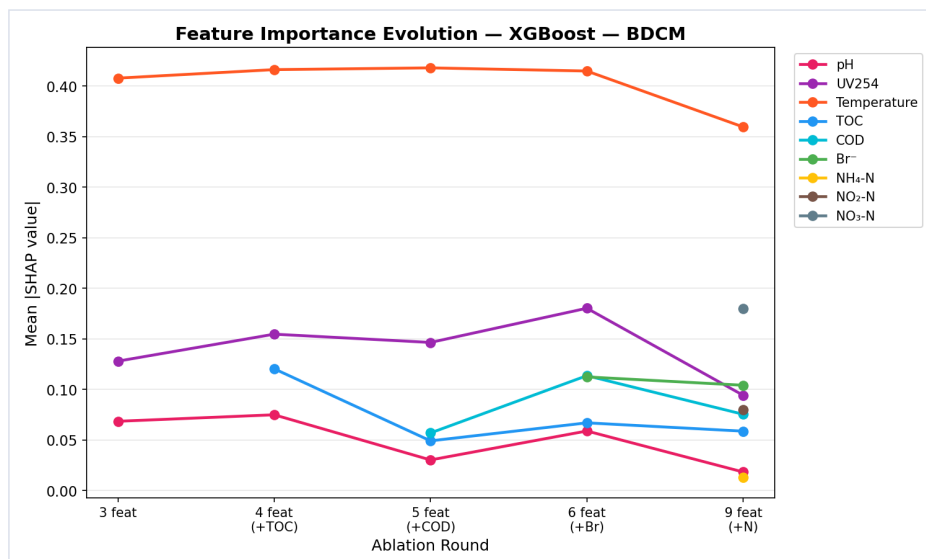


Fig 7c XGBoost importance evolution — BDCM

6 Relative Feature Share (100% Stacked)

How core feature importance redistributes as new features are added

Random Forest — Relative Importance Shift

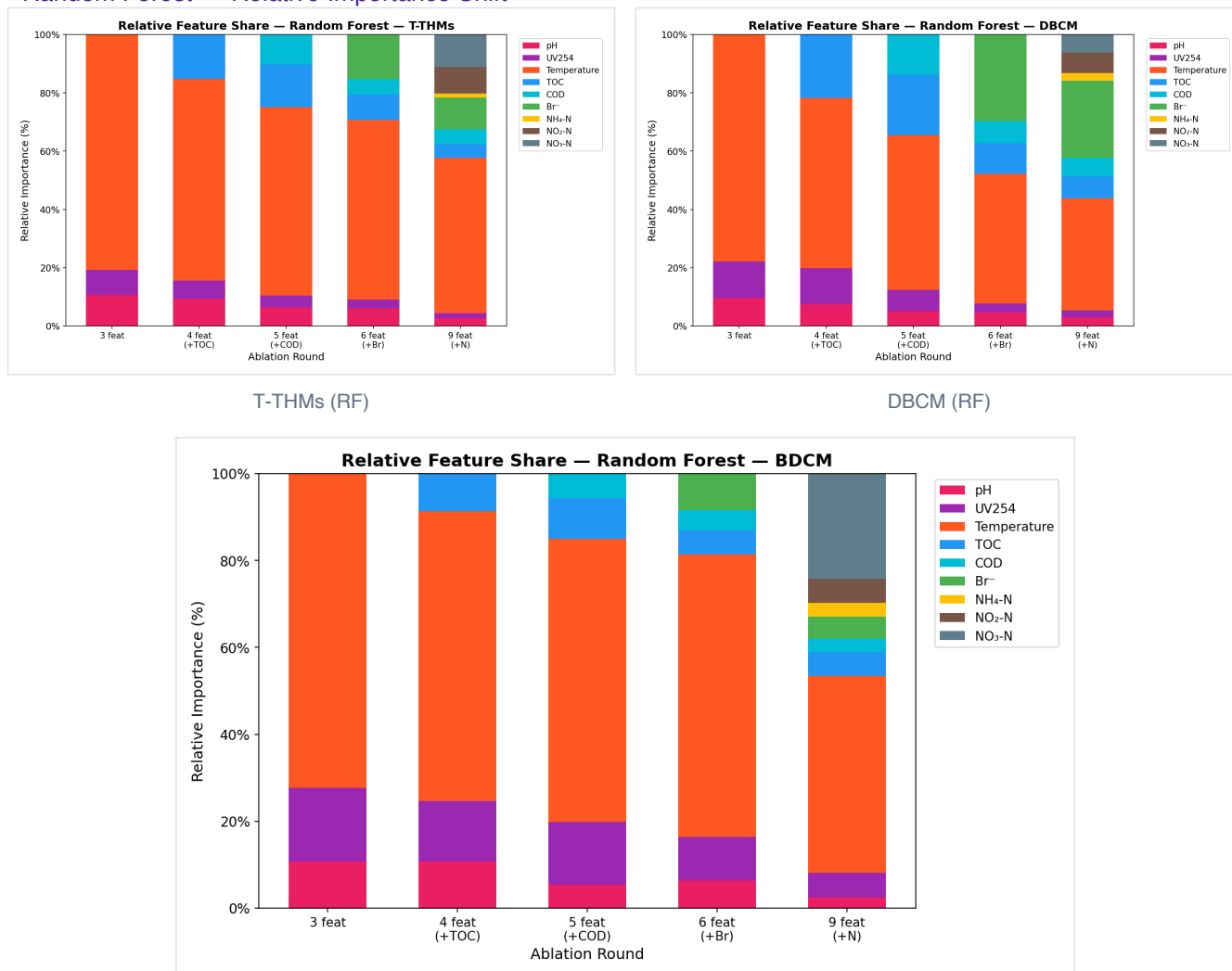


Fig 8c Relative importance — BDCM (RF)

6 Stacked Area — Total Feature Importance

Absolute SHAP stacked area across ablation rounds

Random Forest

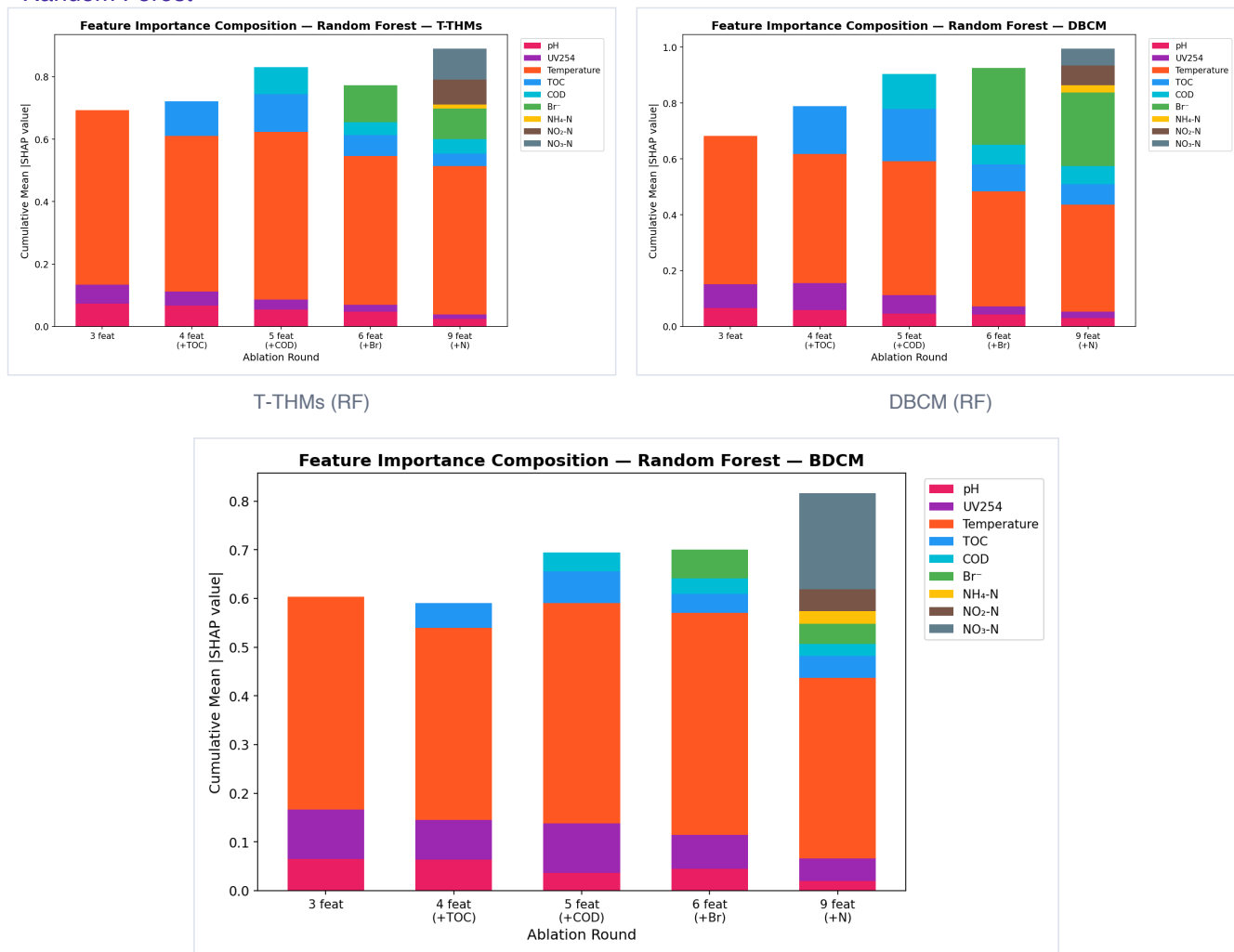


Fig 9c Stacked area — BDCM (RF)

7 Core Feature Displacement — Delta Heatmap

Change in ISHAPI of pH, UV254, Temperature when a new feature is introduced (RF)

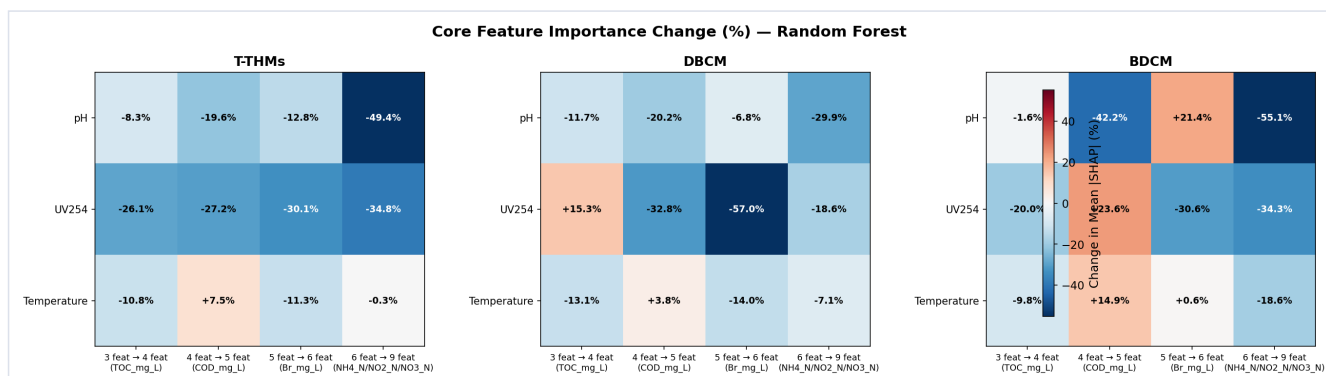


Fig 10 Delta heatmap: $\Delta\text{MeanISHAPI}$ of core features after each addition (RF)

Observations

- Temperature is robust: its SHAP stays relatively stable across rounds — not easily displaced.
- UV254 dilution: loses importance once TOC is introduced (TOC and UV254 are correlated proxies).
- pH: shows minor negative displacement but remains secondary across all rounds.

7 Model Performance vs SHAP Complexity

Test R^2 plotted against total ISHAPI — diminishing returns beyond 5–6 features

Random Forest

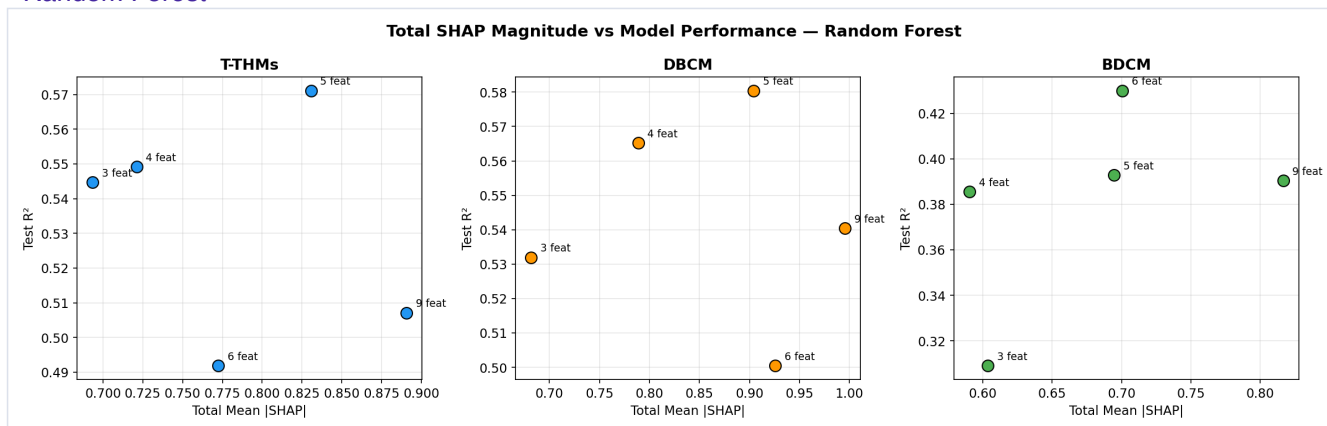


Fig 11a RF: R^2 vs total ISHAPI — peak at 5 features, inflation at 9

XGBoost

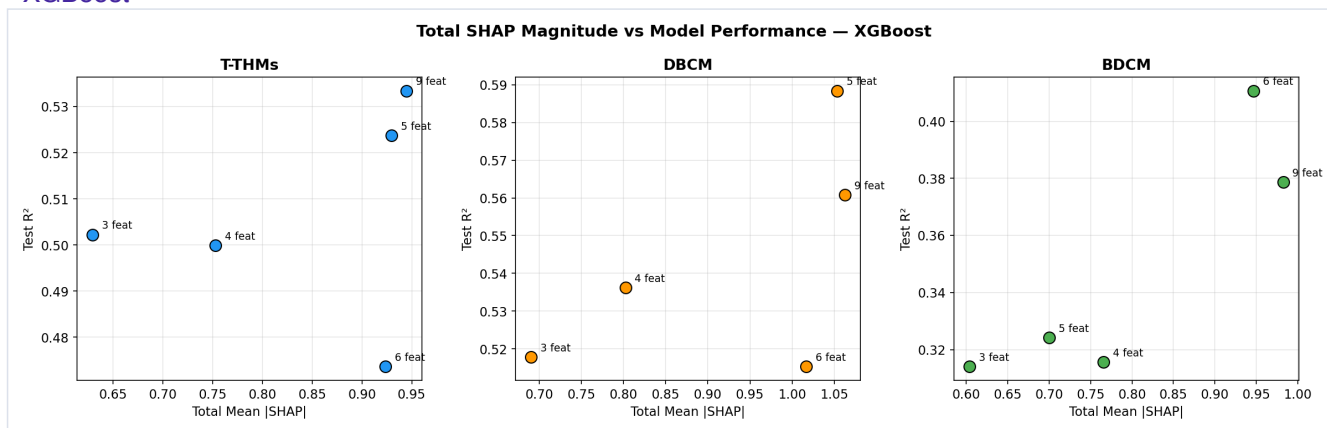


Fig 11b XGBoost: same diminishing-return pattern

8 Neural Network SHAP Evolution

MLP and KAN — importance patterns across ablation rounds

MLP — T-THMs

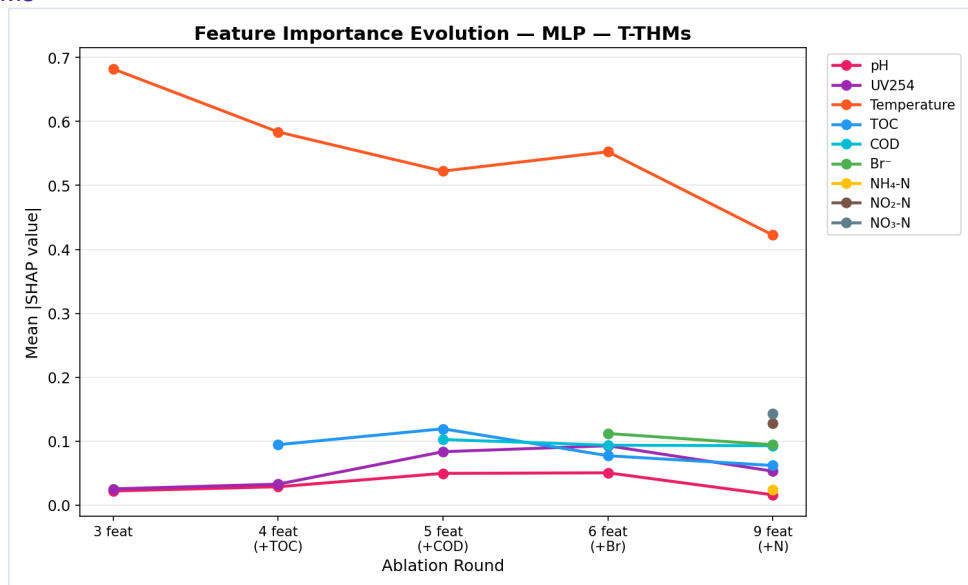


Fig 12a MLP feature importance evolution — T-THMs

KAN — BDCM

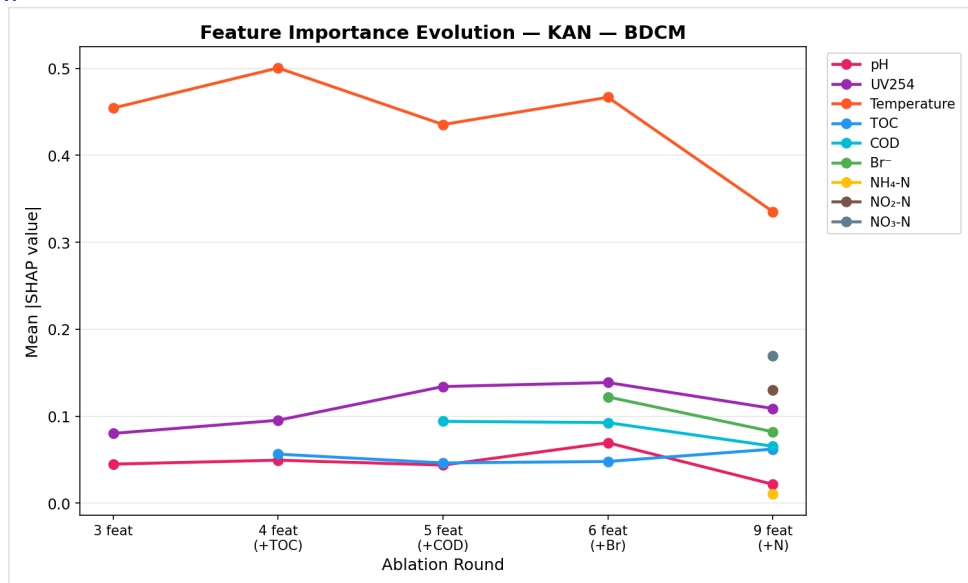


Fig 12b KAN feature importance evolution — BDCM

Observations

- Temperature: dominant in both MLP and KAN; sharper concentration than tree models.
- KAN $\leftrightarrow$ NO₃-N: KAN assigns notable importance to NO₃-N for BDCM at 9-feat (unique to KAN).
- Consensus: all 4 architectures agree — Temperature $\gg$ TOC/COD $\gg$ remaining features.

8 SHAP Beeswarm Grid — Random Forest

Feature distributions per ablation round, all three targets

Target: T-THMs

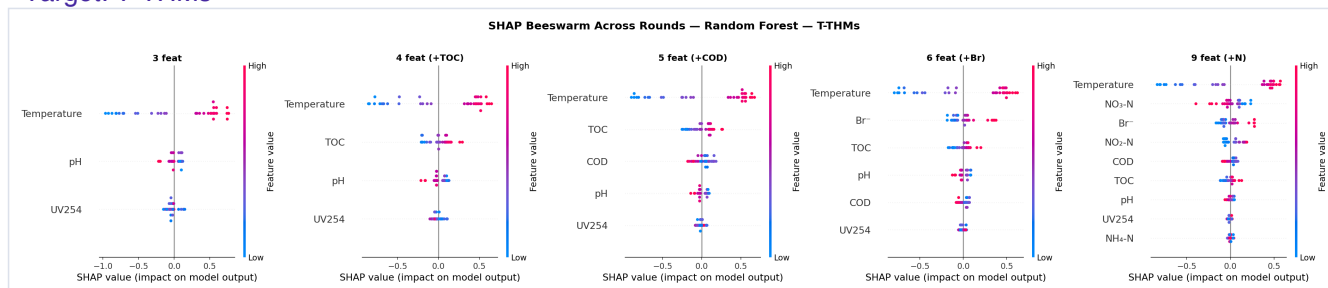


Fig 13a RF beeswarm grid — T-THMs

Target: DBCM

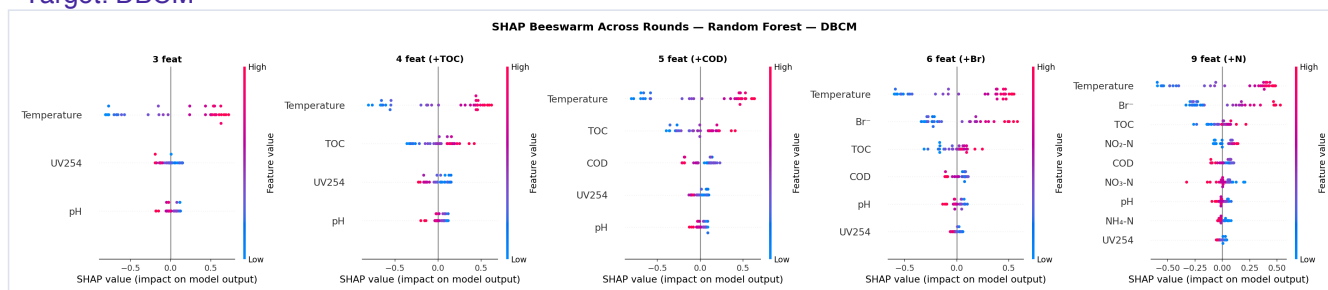


Fig 13b RF beeswarm grid — DBCM

Target: BDCM

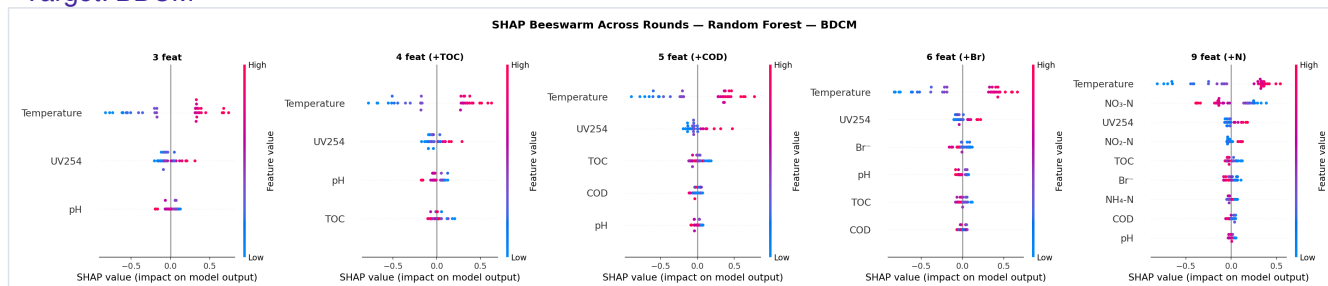


Fig 13c RF beeswarm grid — BDCM

9 Conclusions & Recommendations

Summary of findings and proposed next steps

Optimal Feature Sets per Target

Target	Best Feature Set	Model	R ²	Key Driver
T-THMs	pH + UV254 + Temp + TOC + COD	RF	0.571	Temperature
DBCM	pH + UV254 + Temp + TOC + COD	MLP	0.659	Temp + TOC
BDCM	pH + UV254 + Temp + TOC + COD	MLP	0.434	Temp + Br

Key Findings

- 1. Temperature is dominant: accounts for 55–80% of total SHAP across all models and targets.
- 2. 5-feature optimum: Peak performance for T-THMs and DBCM — extra features add noise not signal.
- 3. Br is target-specific: Essential for BDCM; harmful (dilution effect) for chlorinated DBPs.
- 4. Nitrogen irrelevant: $\text{NH}_4\text{-N}$, $\text{NO}_2\text{-N}$, $\text{NO}_3\text{-N}$ deliver no measurable gain on DWTP-B.
- 5. Cross-model consensus: All 4 architectures agree on the top feature ranking.
- 6. BDCM bottleneck: All models plateau at $R^2 \approx 0.43$; may indicate data limitations or noise.

Recommended Next Steps

- SHAP interaction values: quantify Temperature × TOC and Temperature × Br synergies.
- Cross-plant validation: test the 5-feature model on DWTP-A data.
- Feature engineering: Temp × UV254 interaction term, $\log(\text{TOC})$, Br/TOC ratio.
- Collect more BDCM samples to improve model resolution for brominated DBPs.
- Ensemble: stack RF + XGBoost predictions as a simple performance upper bound.